WATER SOFTENER & MID-PLATE STARTUP PROCEDURE

V. 10/22/2026

STEP 1

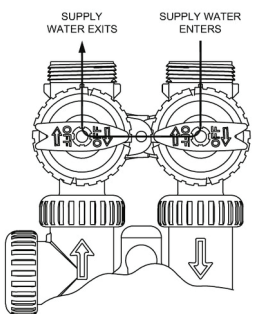
Before Startup

1.) Verify incoming water pressure does not exceed 80 psi.
2.) Verify the inlet and outlet connections are correct and secure.
3.) Verify the drain line is connected, secured, and unobstructed.
4.) Verify the brine line is properly connected.
5.) Verify the brine tank overflow connection is installed where required.
6.) Confirm the system is in Bypass.
7.) Verify electrical power is available.

→ **IMPORTANT:** Inspect plumbing, drain, and brine connections for leaks throughout startup.

Figure 2

BYPASS OPERATION



BYPASS OPERATION
both handles are positioned toward the center of the bypass.

STEP 2

Restore Water & Flush

1.) Open a downstream cold-water faucet.
2.) Slowly open the building's main water supply.
3.) Flush until the water runs clear and all air is expelled.
4.) Close the faucet.

STEP 3

Power & Verify Valve

1.) Plug the system into an electrical outlet.
2.) Pull the battery tab located behind the faceplate.
3.) Verify the valve is properly programmed.
4.) Verify the correct time of day.

CAUTION: Do not remove the battery tab before plugging the valve into an electrical outlet. Valve programming may be damaged if the battery tab is removed before the valve is plugged into an electrical outlet.

STEP 4

Initiate Manual Regeneration

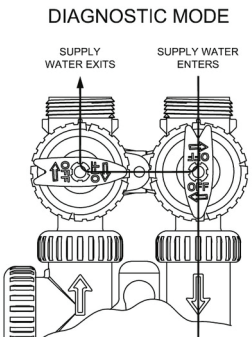
1.) Press and hold REGEN to initiate regeneration.
2.) Allow the valve motor to stop at each cycle position.
3.) Press REGEN after each motor stop to advance through the regeneration sequence.
4.) Continue until the valve reaches the second Backwash cycle.

STEP 5

Fill & Purge Mineral Tank

1.) At the second Backwash, unplug the valve to hold it in Backwash.
2.) Crack the inlet side of the bypass valve approximately 1/8 turn.
3.) Allow the mineral tank to fill slowly.
4.) As the air clears, gradually open the inlet farther.
5.) Continue until the inlet is fully open and full backwash flow is established.
6.) Verify steady flow to drain.
7.) Purge trapped air from the mineral tank by briefly closing the inlet valve. Leave the inlet valve closed for at least 1 minute, then slowly reopen it to the Normal Operation position. Repeat as necessary until air is expelled.
8.) Continue backwashing until the discharge is clear and free of air.
9.) Inspect the system and drain connections for leaks.

Figure 3



DIAGNOSTIC MODE
Inlet open and outlet closed. During initial filling, begin with the inlet approximately 1/8 turn open and gradually open it fully.

STEP 6

Complete Rinse

1.) Plug the control valve back into power.
2.) Press REGEN to advance the valve to Rinse.
3.) Allow the complete 4-minute Rinse cycle to finish.

STEP 7

Add Salt

IMPORTANT: Use solar salt only.

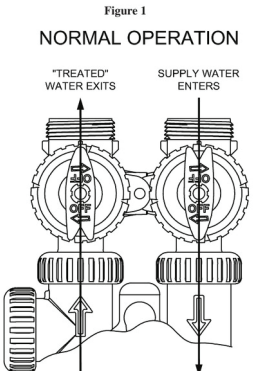
1.) Add high-quality solar salt to the brine tank, filling it approximately 1/2 to 2/3 full.
2.) Verify that salt is not interfering with the brine well, safety float, or brine line.
3.) Replace the brine tank lid.

STEP 8

System Verification

1.) Verify the control valve is in Normal Operation and displays the correct time of day.
2.) Inspect the system and all plumbing connections for leaks.
3.) Confirm the installation area is clean and the system is ready for normal operation.
4.) Test the treated water and verify the system is producing soft water.
5.) Operate all plumbing fixtures to verify proper water flow and system operation.
6.) Review system operation and basic maintenance with the customer, answer any questions, and turn the system over to the customer.

Figure 1



NORMAL OPERATION
Inlet and outlet handles are positioned in the direction of flow.

STEP 9

NOTE: Use only plain, unscented household chlorine bleach. Do Not use scented, splashless, gel, or bleach containing additional cleaning additives.

Sanitize System

1.) Add 1.5 fl oz (45 mL) of plain, unscented 5.25% household chlorine bleach per cubic foot of resin to the brine well.
2.) Press REGEN button 1 time. This sets valve to regenerate at 2:00 AM overnight.